ARTIFICIAL INTELLIGENCE

CBSE Previous Year Question Paper 2023 • Q.P. Code 104 • Set 4
Complete Answer Key: MCQs, Short Answers & Long Answers

Revision note: The paper instructs students to attempt selected questions only. This key answers all visible questions for practice. Equivalent accurate wording is acceptable.

SECTION A — OBJECTIVE TYPE QUESTIONS (MCQs)

Q1. Multiple-choice questions

1(i) (a) Sharing of wealth — this is not generally considered a core task of an entrepreneur; entrepreneurs create value, meet customer needs and may contribute to society.

1(ii) (b) Graphical User Interface — GUI stands for Graphical User Interface.

1(iii) (b) Innovation — entrepreneurs develop or improve products, services and processes.

1(iv) (b) Shortcut Menu — right-clicking a file or folder opens a context/shortcut menu.

1(v) (d) All of the above — stress management can improve mood, support physical health and contribute to longevity.

1(vi) (b) Self-motivation — the inner drive to pursue goals without external pressure or rewards.

Q2. Multiple-choice / fill-in questions

2(i) Pocket assistants include **Siri** and **Google Assistant** (Alexa is another valid example).

2(ii) Any two: linguistic intelligence, logical-mathematical intelligence, spatial intelligence, interpersonal intelligence, intrapersonal intelligence, musical intelligence, bodily-kinaesthetic intelligence, naturalistic intelligence, existential intelligence.

2(iii) (d) (ii) and (iii) — “Data science is one of the domains of AI” and “price comparison websites are examples of data science” are the incorrect statements in the question’s intended framing.

2(iv) (a) Training Data — data used to train or fit the model is called training data.

2(v) False. Regression predicts a continuous numerical value; classification assigns labels or categories.

2(vi) (b) F1 Score — it is the harmonic mean of precision and recall.

Q3. Multiple-choice questions

3(i) (c) NLP — email filters, spam filters and smart assistants are applications involving Natural Language Processing, although some may also use other AI techniques.

3(ii) (d) All of the above — smart bots can be flexible, require implementation/coding, and may connect to large databases or other resources.

3(iii) (b) Sentence Segmentation — it splits a text corpus into individual sentences.

3(iv) Evaluation — it assesses how well a model represents the data and is expected to perform on new data.

3(v) (b) (ii) and (iii) — recall uses true positives and false negatives: TP/(TP+FN).

3(vi) (a) Rare / valuable words — in the graph, X is at the low-occurrence end, representing infrequent words that may carry useful information.

Q4. Multiple-choice / fill-in questions

4(i) (a) Helps in classifying the type and genre of a document.

4(ii) True positive and **true negative** — these occur when a model’s prediction agrees with reality.

4(iii) (c) They are modelled on the human brain and nervous system.

4(iv) Computer Vision — CV stands for Computer Vision.

4(v) Problem Scoping — the problem is examined from different perspectives and its key elements are identified.

4(vi) Supervised learning — the training dataset is labelled.

Q5. Multiple-choice / fill-in questions

5(i) Token — a token is a word, number or special character treated as a unit during text processing.

5(ii) True positive or true negative, depending on the class being evaluated. A prediction that matches reality is a correct prediction; the question likely expects “True Positive” in the positive-class context.

5(iii) (a) Natural Language Processing — virtual assistants such as Alexa and Siri interpret spoken language using NLP.

5(iv) (a) Problem Scoping — the 4Ws Problem Canvas is used while defining and understanding the problem.

5(v) (c) Clustering — an unsupervised method that groups data points based on similarities or patterns.

5(vi) (a) Accuracy — it measures the proportion of all predictions that are correct.

SECTION B — SUBJECTIVE TYPE QUESTIONS

Short Answers

Q6–Q10: Employability Skills

Q6. How does meditation help in managing stress? Discuss briefly.

Meditation calms the mind, supports focused breathing and reduces anxious thoughts. Regular practice can improve concentration, emotional control and the ability to respond calmly to stressful situations.

Q7. Give any two key roles performed by an entrepreneur.

An entrepreneur identifies business opportunities and organises resources to start a venture. An entrepreneur also takes calculated risks, makes decisions and coordinates people and activities to deliver products or services.

Q8. Mention any two benefits of working independently.

Working independently builds self-confidence and responsibility. It also improves decision-making, time management and the ability to complete tasks without constant supervision.

Q9. Suggest two points to prevent a new office computer from virus infection.

Install reputable antivirus software and keep it updated. Keep the operating system and applications updated, and avoid opening suspicious email attachments or downloading files from untrusted websites.

Q10. Define agricultural entrepreneurship. How do farmers benefit from it?

Agricultural entrepreneurship means starting or managing a business based on farming and related activities, such as food processing or farm services. It can increase farmers’ income, create local jobs, add value to produce and connect farmers with wider markets.

Q11–Q16: Artificial Intelligence

Q11. Explain one example of AI bias.

If a hiring AI is trained mainly on past data from one demographic group, it may unfairly rank applicants from other groups lower. This is bias caused by unrepresentative training data.

Q12. What is Dimensionality Reduction?

Dimensionality reduction is the process of reducing the number of input features in a dataset while preserving as much useful information as possible. It can simplify models and reduce computation.

Q13. Define Chatbot. What are its types?

A chatbot is software that communicates with users through text or voice. Common types are rule-based/script bots, which follow predefined rules, and AI-powered smart bots, which interpret input more flexibly.

Q14. Define Confusion Matrix.

A confusion matrix is a table comparing a classification model’s predictions with actual outcomes. It summarises true positives, true negatives, false positives and false negatives.

Q15. Explain smartphone face lock as an example of Computer Vision.

Face unlock uses a camera to capture facial features and computer-vision algorithms to compare them with the enrolled face pattern. If the match passes the security threshold, the phone unlocks.

Q16. Expand TF-IDF and give two applications.

TF-IDF means Term Frequency–Inverse Document Frequency. It weights a word according to how frequent it is in one document and how uncommon it is across a collection. Applications include search-engine ranking and document/information retrieval.

LONG ANSWERS (50–80 WORDS EACH)

Q17. Differentiate AI, ML and DL. Include a labelled diagram.

Artificial Intelligence (AI) is the broad field of building systems that perform tasks requiring capabilities such as reasoning or perception. Machine Learning (ML) is a subset of AI in which systems learn patterns from data. Deep Learning (DL) is a subset of ML that uses neural networks with multiple layers. Thus, DL is within ML, and ML is within AI.

Artificial Intelligence (AI)
■■■ Machine Learning (ML)
■■■ Deep Learning (DL)

Q18. Explain the significance of the AI Project Cycle and distinguish Data Acquisition from Data Exploration.

The AI Project Cycle provides a structured process for solving problems with AI: Problem Scoping, Data Acquisition, Data Exploration, Modelling and Evaluation. Data Acquisition means collecting relevant data from sources such as surveys, sensors or databases. Data Exploration means examining and visualising the collected data to understand its structure, quality, patterns and missing values before building a model.

Q19. Create a Bag of Words document vector table for the two documents.

Step 1 — lowercase and remove punctuation. Doc 1: “sameera and sanya are classmates”. Doc 2: “sameera likes dancing but sanya loves to study mathematics”. Step 2 — tokenise into words. Step 3 — create a vocabulary in first-appearance order. Step 4 — count each vocabulary term in each document. The count vectors below form the Bag of Words representation.

Vocabulary	Document 1	Document 2
sameera	1	1
and	1	0
sanya	1	1
are	1	0
classmates	1	0
likes	0	1
dancing	0	1
but	0	1
loves	0	1
to	0	1
study	0	1
mathematics	0	1

Q20. Are all devices called “smart” AI-enabled? Give two examples of devices commonly misunderstood as AI.

No. “Smart” often means a device has digital controls or connectivity; it does not automatically mean it uses AI. A programmable digital alarm clock follows set rules, and a basic automatic washing machine may run fixed cycles. These devices can operate without learning from data or making AI-based predictions, so they are not necessarily AI-enabled.

Q21. Earthquake prediction confusion matrix: calculate TN, precision, recall and F1-score.

Taking “Yes” (earthquake predicted/actual) as the positive class: TP = 50, FP = 5, FN = 25 and TN = 20. True Negatives = **20**. Precision = TP/(TP+FP) = 50/55 = **90.91%**. Recall = TP/(TP+FN) = 50/75 = **66.67%**. F1-score = 2TP/(2TP+FP+FN) = 100/130 = **76.92%**.

Notes for revision

• Q2(iii): The question’s wording is somewhat ambiguous. The answer above follows the likely intended school-level classification of statements (ii) and (iii) as incorrect; revise the domain descriptions in your prescribed textbook.
• Q5(ii): A correct prediction can be a true positive or true negative depending on whether the case is positive or negative. The paper’s fill-in wording is underspecified without the class context.
• Q21: Values are read from the matrix on page 7: predicted Yes/actual Yes = 50; predicted Yes/actual No = 5; predicted No/actual Yes = 25; predicted No/actual No = 20.

Prepared from the uploaded 2023 question paper, including the Bag of Words prompt and confusion matrix shown on page 7.